



# Modelling Financial Markets

## Seminar 13: Machine Learning in Finance — Hands-on



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## Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ CI: Confidence Interval
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation
- ▣ FFN: FeedForward Network
- ▣ GAN: Generative Adversarial Network
- ▣ GB: Gradient Boosting
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IS: In-Sample
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LS: Linear SDF estimated by unregularised Least Squares
- ▣ LW: Local Whittle (estimator)
- ▣ MDA: Mean Decrease Accuracy
- ▣ MDI: Mean Decrease Impurity
- ▣ OLS: Ordinary Least Squares

## Acronyms used in this material (2/2)

- ▣ OOS: Out-Of-Sample
- ▣ PBO: Probability of Backtest Overfitting
- ▣ PSR: Probabilistic Sharpe Ratio
- ▣ PT: Pesaran–Timmermann (test of directional accuracy)
- ▣ RC: Reality Check (White's test for data snooping)
- ▣ RF: Random Forest
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ SD: Standard Deviation
- ▣ SDF: Stochastic Discount Factor
- ▣ SE: Standard Error
- ▣ SHAP: SHapley Additive exPlanations
- ▣ SPA: Superior Predictive Ability (test)
- ▣ SR: Sharpe Ratio
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index

## Today's Question and Route

- ▣ Question of the seminar: **does any machine-learning signal on Bitcoin survive honest labels, honest validation and honest selection?**
- ▣ Route
  - ▶ First the primer “What You Need for Today” (five slides): [Chapter 13](#) comes after the seminar
  - ▶ Part I: memory (A1), labels (A3, B3), validation (A7, B5)
  - ▶ Part II: prediction and its uncertainty (B7a, B7b, B8), an AI answer to critique (C2), selection and deflation (A5)
  - ▶ After the lecture: B1 (local Whittle memory of Bitcoin) and B10 (grid search and the effective number of trials)
- ▣ What you produce: the A answers, one chart and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- ▣ Seminar: Antoaneta Amza, PhD candidate, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

## Exercise Map

| Exercise          | Question                                                                                                         | Type     | Model              |
|-------------------|------------------------------------------------------------------------------------------------------------------|----------|--------------------|
| A1                | what does a truncated fractional filter keep of the price level?                                                 | Solved   | –                  |
| A3, B3            | how do triple-barrier labels behave on toy paths and on Bitcoin?                                                 | Solved   | –                  |
| A7, B5            | what does purging remove, and what does shuffling fake?                                                          | Solved   | –                  |
| B7a, B7b          | do three models beat “always up”, and how uncertain is the AUC?                                                  | Solved   | –                  |
| B8                | does a model forecast better than a constant?                                                                    | Proposed | B7a, B7b           |
| C2                | what is wrong in an AI answer?                                                                                   | Proposed | B5, B7a            |
| A5                | is the best of 100 backtests better than luck?                                                                   | Solved   | –                  |
| After the lecture | B1: does fractional differencing make Bitcoin stationary?                                                        | Solved   | –                  |
| Extension packs   | B10: the best of 120 rules against luck<br>A2, A3 Extended, A4, A6, A8–A10, B2, B4, B6, B7c, B9, B11–B14, C1, C3 | both     | named on each task |

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations and small numerical cases; Part B: verification and inference on crypto data, each ending with an interpretation question; Part C: an AI critique and project seeds

## Data Used

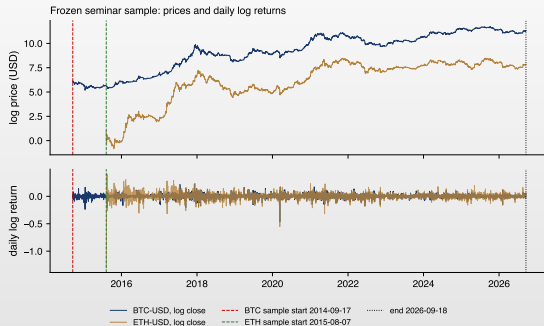
| Series                                | Field                 | Frozen sample                                           | Used in            |
|---------------------------------------|-----------------------|---------------------------------------------------------|--------------------|
| Bitcoin                               | close                 | 2014-09-17 to 2026-09-18,<br>4,385 closes               | B1, B3–B14, C1, C2 |
| Ether                                 | close                 | 2015-08-07 to 2026-09-18,<br>4,061 closes               | B2                 |
| VIX                                   | close                 | US trading days, carried<br>forward to the BTC calendar | B14                |
| Kenneth French<br>factors and deciles | Data Library, monthly | 1967 – latest month                                     | C3                 |

- Daily closes of the course data, one series per asset (field used: close); crypto trades every day, so annualise with  $q = 365$
- Returns: daily log returns  $r_t = \ln(P_t/P_{t-1})$  on each asset's own calendar; no missing closes in the frozen samples; events without a complete  $h$ -day window at the end are not labelled

## Setup: Run and Verify

```
# notebook section 0 defines load_data
btc = load_data('btc')['Close']
eth = load_data('eth')['Close']
print(len(btc), btc.index[0], btc.index[-1])
print(len(eth), eth.index[0], eth.index[-1])
r = np.log(btc).diff().dropna()
print(round(r.std() * np.sqrt(365), 3))
```

- Expected: BTC 4,385 closes, 17 Sep 2014 (457.33) to 18 Sep 2026 (80,901.46), annualised volatility 0.666;
- ETH 4,061 closes from 7 Aug 2015
- If a count differs, stop and check the series, the field and the dates



## Notation and the Supplied Toolbox

- $P_t$ : close on day  $t$ ;  $x_t = \ln P_t$ ;  $r_t = x_t - x_{t-1}$ ;  $B$ : backshift,  $Bx_t = x_{t-1}$ ; decision time: the close of day  $t$ , trades at the next close
- $d$ : order of a fractional filter (a choice);  $\delta$ : memory of a series,  $I(\delta)$  (an unknown parameter we estimate)
- $\sigma_{t_0}$ : EWMA volatility of daily log returns, span 50;  $h$ : label horizon;  $[t_0, t_1]$ : the span of a label
- SR: per-day Sharpe ratio  $\bar{r}/\hat{\sigma}$  of daily log returns (risk-free rate 0), annualised  $\times \sqrt{365}$ ;  $\gamma_3$  skewness,  $\gamma_4$  ordinary kurtosis (Normal distribution: 3)

| Supplied function (notebook section 0)                                                                                               | What it returns                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ffd_weights(d, threshold),<br/>frac_diff_ffd</code>                                                                            | weights $w_0, \dots, w_K$ ; the filtered series from day $K$ on                                                                                          |
| <code>local_whittle,<br/>exact_local_whittle</code>                                                                                  | $(\hat{\delta}, \text{SE})$ with $m = \lfloor n^{0.65} \rfloor$ frequencies                                                                              |
| <code>get_daily_vol, triple_barrier<br/>PurgedKFold, build_features<br/>probabilistic_sharpe_ratio,<br/>deflated_sharpe_ratio</code> | EWMA volatility; one row per event: $t_1$ , return, label, exit type<br>purged and embargoed folds; the 11 features of B7<br>PSR and DSR of a per-day SR |

## What You Need for Today (1/5): Classifiers and Their Scores

- Chapter 13 comes after this seminar: these five slides are the minimum for today's core tasks; depth and evidence in the lecture
- Set-up: features  $x_t$  known at the close of day  $t$ , label  $y_t \in \{0, 1\}$  (up or down over the next days); a classifier returns  $\hat{p}_t = \hat{P}(y_t = 1 \mid x_t)$  and predicts “up” when  $\hat{p}_t > 0.5$
- The three models of B7a
  - ▶ Logit-L1:  $\hat{p} = 1/(1 + e^{-x^\top \beta})$  with the penalty  $\lambda \|\beta\|_1$ , which sets weak coefficients exactly to zero (Tibshirani (1996)); in scikit-learn  $C = 1/\lambda$ , so a small  $C$  is a strong penalty
  - ▶ Random forest (RF): the average of many decision trees, each grown on a bootstrap sample with a random subset of features at every split (Breiman (2001))
  - ▶ Gradient boosting (GB):  $F_m = F_{m-1} + \eta h_m$ , each small tree  $h_m$  corrects what the previous trees missed;  $\eta$  is the learning rate (Friedman (2001))
- Accuracy: share of correct classes; its benchmark is “always up”, whose accuracy is the share of up labels, not 50%
- AUC (area under the ROC curve, Hanley & McNeil (1982)):  $P(\hat{p}_+ > \hat{p}_-) + \frac{1}{2}P(\hat{p}_+ = \hat{p}_-)$ , the chance that a random up day gets a higher score than a random down day; 0.5 = no ranking
  - ▶ Example: up days scored (0.7, 0.4), down days (0.5, 0.2); of the 4 pairs, 3 are ordered correctly (0.4 < 0.5 is not): AUC = 0.75

## What You Need for Today (2/5): Losses and Their Uncertainty

- Out-of-fold prediction: every day is predicted by a model that was not trained on it (slide 4/5 says which days must also be left out)
- Log loss  $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$ : penalises confident wrong probabilities; AUC judges the ranking, log loss the calibration (B8)
  - ▶ Example:  $y_t = 1$ ;  $\hat{p}_t = 0.6$  costs  $-\ln 0.6 = 0.51$ ,  $\hat{p}_t = 0.2$  costs  $-\ln 0.2 = 1.61$
- Comparing two forecasts: loss differential  $d_t = \ell_t^A - \ell_t^B$  and the Diebold–Mariano statistic  $\bar{d}/\widehat{SE}(\bar{d})$  (Diebold & Mariano (1995)), with Newey–West errors (Newey & West (1987)) because neighbouring  $d_t$  are dependent
- Standard error of an AUC
  - ▶ DeLong (DeLong, DeLong & Clarke-Pearson (1988)): a closed-form SE that treats the days as independent
  - ▶ Circular block bootstrap (Künsch (1989)): resample blocks of  $L$  consecutive days (wrapping around the end), recompute the AUC, read the SE and the percentile CI from the replications;  $L = 1$  is the ordinary bootstrap
  - ▶ Blocks keep the dependence inside them, so with overlapping labels the block SE is the honest one (B7b)

## What You Need for Today (3/5): Fractional Differencing

- Differencing of order  $d$  with the backshift  $B$ :  $(1 - B)^d x_t = \sum_{k \geq 0} w_k x_{t-k}$ ,  $w_0 = 1$ ,  $w_k = -w_{k-1} \frac{d-k+1}{k}$  (Hosking (1981));  $d = 0$  keeps the level,  $d = 1$  gives the return
  - ▶ Example  $d = 0.5$ :  $w = (1, -0.5, -0.125, -0.0625, -0.039, \dots)$ ; the weights decay slowly, so a fractional  $d$  keeps part of the old levels: “memory”
- FFD (fixed-width window fractional differencing, López de Prado (2018)): keep the weights until the first  $|w_k| < \tau$  and apply the same  $K + 1$  weights every day
- Memory  $\delta$ :  $x_t$  is  $I(\delta)$  when  $(1 - B)^\delta x_t$  has short memory
  - ▶  $\delta < \frac{1}{2}$ : stationary;  $\frac{1}{2} \leq \delta < 1$ : non-stationary but mean-reverting;  $\delta = 1$ : unit root, a random walk; the local Whittle estimate of  $\delta$  is in Chapter 1
  - ▶ Filtering an  $I(1)$  series with order  $d$  leaves memory  $1 - d$ , if the filter is not truncated
- The minimum- $d$  rule (López de Prado (2018)): the smallest  $d$  at which the ADF test (Dickey & Fuller (1979)) rejects; ADF:  $H_0$  unit root, reject when the statistic is below the critical value ( $-2.86$  at 5% with a constant); A1 asks what such a rejection proves

## What You Need for Today (4/5): Labels and Honest Validation

- Fixed-horizon label:  $y_t = 1\{\ln P_{t+h} - \ln P_t > 0\}$ ; volatility scale  $\sigma_{t_0}$ : EWMA standard deviation of daily log returns, span 50 (weight  $2/51 \approx 0.039$  on the newest day)
- Triple barrier (López de Prado (2018)): from the entry  $t_0$ , barriers  $P_{t_0} e^{+pt \sigma_{t_0} \sqrt{h}}$ ,  $P_{t_0} e^{-sl \sigma_{t_0} \sqrt{h}}$  and a vertical one at  $t_0 + h$ ; label  $+1 / -1$  by the first horizontal touch, else the sign of the return at  $t_0 + h$ ;  $t_1$  = exit day
  - ▶ Example:  $P_{t_0} = 100$ ,  $\sigma = 2\%$ ,  $h = 4$ ,  $pt = sl = 1$ : width  $0.02\sqrt{4} = 0.04$ , barriers 104.08 and 96.08; closes 101, 97, 95: lower touch on day 3, label  $-1$
- Labels of neighbouring days overlap: the label spans  $[t_0, t_1]$  share returns, so the days are not independent
- $K$ -fold cross-validation: train on  $K - 1$  folds, test on the remaining one; **shuffled** folds mix days, **contiguous** folds keep blocks of time
- **Purging** (López de Prado (2018)): drop from training every observation whose label span overlaps a test span; **embargo**: also drop the  $e$  observations right after the test fold
  - ▶ Example:  $n = 10$ , spans  $[i, i + 2]$ , test fold  $\{4, 5\}$  (labels up to day 7),  $e = 1$ : drop  $\{2, 3\}$  before and  $\{6, 7\}$  after; train on  $\{0, 1, 8, 9\}$

## What You Need for Today (5/5): Selection and the Deflated Sharpe Ratio

- Probabilistic Sharpe ratio (PSR, [Bailey & López de Prado \(2012\)](#)): the probability that the true per-day SR exceeds a benchmark  $SR^*$ , given  $T$  returns with skewness  $\gamma_3$  and kurtosis  $\gamma_4$ 
  - ▶  $PSR(SR^*) = \Phi((\widehat{SR} - SR^*)\sqrt{T-1}/D)$ ,  $D = \sqrt{1 - \gamma_3\widehat{SR} + \frac{\gamma_4-1}{4}\widehat{SR}^2}$  ( $D = 1$  to first order for Normal returns)
- Selection: the best of  $N$  strategies with zero true skill still shows a positive SR; its expected value is
  - ▶  $SR_0 = \sqrt{V} \left[ (1 - \gamma_E)\Phi^{-1}(1 - \frac{1}{N}) + \gamma_E\Phi^{-1}(1 - \frac{1}{N_E}) \right]$ ,  $V$  the variance of the SR across trials,  $\gamma_E = 0.5772$  Euler's constant
- Deflated Sharpe ratio (DSR, [Bailey & López de Prado \(2014\)](#)):  $DSR = PSR(SR_0)$ , the PSR against the luck of the best trial
- Example: per-day  $\widehat{SR} = 0.10$ ,  $T = 400$ ,  $D = 1$ :  $PSR(0) = \Phi(0.10\sqrt{399}) = \Phi(2.00) = 0.977$ 
  - ▶ If it is the best of  $N = 10$  trials with  $V = 1/T$ :  $SR_0 = \frac{1}{20}[0.4228 \cdot 1.282 + 0.5772 \cdot 1.789] = 0.079$ ,  
 $DSR = \Phi((0.10 - 0.079)\sqrt{399}) = \Phi(0.43) = 0.66$
- More in [Chapter 13](#), after the seminar: the false strategy theorem, feature importance, walk-forward backtests and the S&P 500 case study

## A1 [Solved]: FFD Weights and the Retained Window — Task

- ▣ **Question:** what does a truncated fractional filter keep of a price level?
- ▣ **Inputs:** weights  $w_0 = 1$ ,  $w_k = -w_{k-1} \frac{d-k+1}{k}$  (Hosking (1981)); truncation at the first  $|w_k| < \tau = 10^{-4}$  (López de Prado (2018));  $K$  = the last retained lag, so the window uses  $K + 1$  observations; no data file
- ▣ **Tasks**
  1. Compute  $w_0, \dots, w_4$  for  $d = 0.25$  with the recursion.
  2. With `ffd_weights(0.25)`, report  $K$ , the window length  $K + 1$  and  $\sum_{k \leq K} w_k$ ; repeat for  $d = 0.3$  and  $d = 0.5$ .
  3. Write  $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$  as  $(\sum_k w_k) X_t$  minus a combination of the differences  $X_t - X_{t-k}$ .
  4. Decide whether  $\tilde{X}_t$  of an  $I(1)$  log price is stationary.
- ▣ **Report:** the five weights, a table ( $d$ ,  $K$ ,  $K + 1$ , weight sum), the decomposition and the decision
- ▣ **Interpretation:** the minimum- $d$  rule picks the smallest  $d$  at which an ADF test rejects: what does such a rejection prove?
- ▣ Notebook: section A1

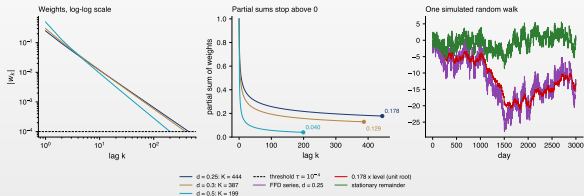
## A1 [Solved]: Solution

1.  $w_1 = -0.25$ ,  $w_2 = -0.25 \cdot 0.75/2 = -0.09375$ ,  
 $w_3 = -0.0547$ ,  $w_4 = -0.0376$ : every weight after  $w_0$  is negative

2. Table:

| $d$  | $K$ | $K+1$ | $\sum_{k \leq K} w_k$ |
|------|-----|-------|-----------------------|
| 0.25 | 444 | 445   | 0.178                 |
| 0.30 | 387 | 388   | 0.129                 |
| 0.50 | 199 | 200   | 0.040                 |

3.  $\tilde{X}_t = (\sum_{k \leq K} w_k) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$
4. Each  $X_t - X_{t-k}$  is  $I(0)$ , but the first term is  $0.178 X_t$  at  $d = 0.25$ :  $\tilde{X}_t$  is still  $I(1)$



- Right panel: one simulated random walk; the scaled level (red) carries the unit root of the FFD series (correlation 0.93)
- Interpretation: an ADF rejection proves that the test confuses a small unit-root component with the stationary part, not that the series is stationary

## A3 [Solved]: Triple-Barrier Labels of Two Paths — Task

### Questions

- ▶ which barrier is touched first?
- ▶ which label results?

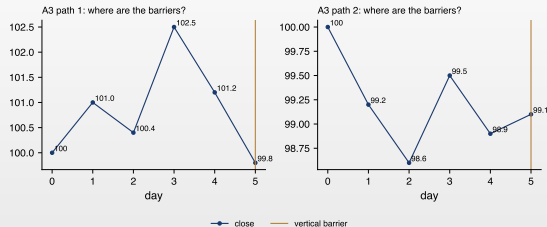
- Inputs: entry  $P_{t_0} = 100$ , daily volatility  $\sigma_{t_0} = 1\%$ , vertical barrier  $h = 5$  days,  $pt = sl = 1$ ; barriers  $P_{t_0} e^{\pm pt \sigma_{t_0} \sqrt{h}}$  (López de Prado (2018)); daily closes only

### Tasks

1. Compute the upper and lower barrier.
2. Draw both on the two paths and mark the first touch.
3. Label each path: +1 upper, -1 lower, and at the vertical barrier the sign of  $\ln(P_{t_0+h}/P_{t_0})$ .

- Report: the two barriers, the exit day and type, the two labels

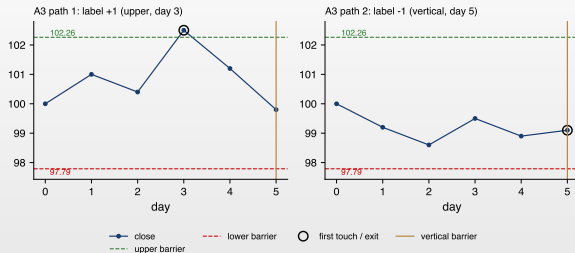
- Interpretation: why can a path that ends below 100 still get +1? Notebook: section A3



## A3 [Solved]: Solution

1. Width  $0.01\sqrt{5} = 0.02236$ : upper  $100e^{0.02236} = 102.26$ , lower  $100e^{-0.02236} = 97.79$
2. Path 1: day 3,  $102.5 \geq 102.26$ , upper touch; path 2: no horizontal touch in 5 days
3. Path 1: +1 (the later fall to 99.8 does not matter); path 2:  $\ln(99.1/100) = -0.90\%$  at the vertical barrier: -1

- Interpretation: the label records the first exit a trader would have taken, not the end-of-window price; a timeout keeps the sign of the terminal return, so “vertical” and “-1” are different facts



## B3 [Solved]: Triple-Barrier Labels on Bitcoin — Task

### ▣ Questions

- ▶ how often does Bitcoin hit a horizontal barrier within 10 days?
- ▶ how balanced are the labels?

- ▣ **Data and settings:** BTC daily closes; every day is an event  $t_0$ ;  $\sigma_{t_0}$  = EWMA volatility, span 50 (`get_daily_vol`);  $h = 10$  days; monitoring on daily closes; events without 10 later closes are dropped (4,373 events, 19 Sep 2014 – 8 Sep 2026)

### ▣ Tasks

1. With `triple_barrier`, label every event for  $pt = sl = 1$  and  $pt = sl = 2$ ; print one event record ( $t_0$ ,  $\sigma_{t_0}$ , barrier prices,  $t_1$ , exit type, return, label).
2. Compute the shares of upper, lower and vertical exits, the average holding period and the share of +1 labels.
3. Cross-tabulate exit type against the final label, and compare with the 10-day fixed-horizon label on the same events.

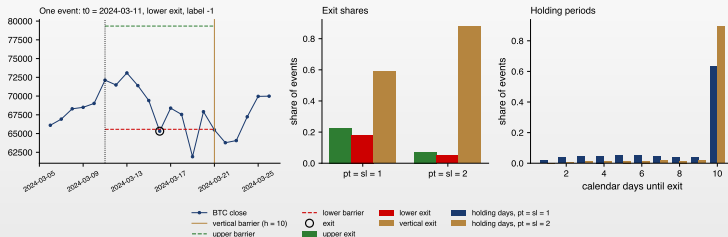
- ▣ **Report:** the event record, a table by setting, the cross-tabulation, the exit-share and holding-period chart

### ▣ Interpretation

- ▶ why are about 56% of the labels +1?
- ▶ what is then the accuracy to beat?

- ▣ Notebook: section B3

## B3 [Solved]: Solution

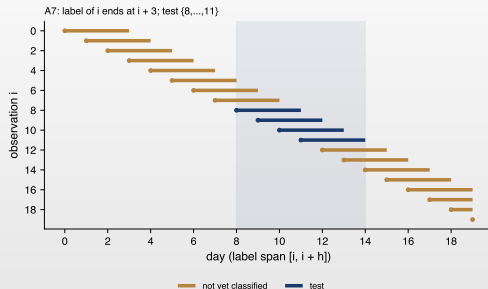


|               | Upper | Lower | Vertical | Days | +1    |
|---------------|-------|-------|----------|------|-------|
| $pt = sl = 1$ | 22.8% | 18.2% | 59.0%    | 8.25 | 55.7% |
| $pt = sl = 2$ | 7.2%  | 4.9%  | 87.9%    | 9.58 | 55.6% |
| fixed 10 days | —     | —     | 100%     | 10   | 55.5% |

- Event 11 Mar 2024: close 72,124,  $\sigma = 3.0\%$ , barriers 79,336 and 65,568; lower exit on 16 Mar, return  $-9.9\%$ , label  $-1$
- $pt = sl = 1$ : vertical exits split into 1,442 labels  $+1$  and 1,139 labels  $-1$ ; horizontal exits give 995  $+1$  and 797  $-1$
- Interpretation: BTC rose over the sample, so “always up” (about 56%) is the benchmark; wider barriers push labels towards fixed-horizon labels

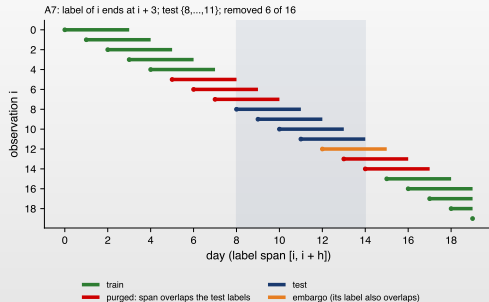
## A7 [Solved]: How Much Does Purging Cost? — Task

- **Question:** which training observations must go when the test labels overlap them?
- **Inputs:**  $n = 20$  observations  $i = 0, \dots, 19$  (zero-based) in  $K = 5$  contiguous folds of  $L = 4$ ; the label of  $i$  spans the closed interval  $[i, i + h]$ ,  $h = 3$ ; embargo  $e = 1$  observation after the last test index; test fold  $\{8, \dots, 11\}$ , whose labels reach day 14
- **Tasks**
  1. List the observations before the test fold whose span reaches day 8 (purged before).
  2. List the embargoed observation and those after the fold that start on or before day 14 (purged after).
  3. Count the union of removals and the share of the 16 non-test observations lost; for an interior fold with  $n - L$  large enough on both sides, show the share is  $\frac{h + \max(h, e)}{n - L}$ .
- **Interpretation:** why does shuffled  $K$ -fold overstate accuracy in this setting? Notebook: section A7



## A7 [Solved]: Solution

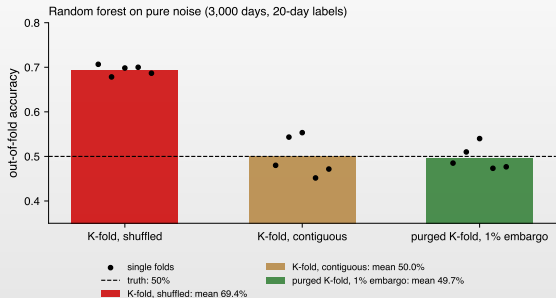
1. Purged before:  $\{5, 6, 7\}$ , since  $i + 3 \geq 8$
  2. Embargo:  $\{12\}$ ; purged after:  $\{12, 13, 14\}$ , since they start by day 14; the embargoed 12 is also purged
  3. Union:  $\{5, 6, 7, 12, 13, 14\}$ , 6 of 16 lost (37.5%); training set  $\{0, \dots, 4\} \cup \{15, \dots, 19\}$
- General interior fold:  $h$  observations before; after, the labels starting before the last test label ends ( $h$ ) or inside the embargo ( $e$ ):  $\max(h, e)$ ; share  $\frac{h + \max(h, e)}{n - L} = \frac{3 + 3}{16}$
  - Interpretation: shuffling places neighbours that share returns on both sides of the split; a flexible model memorises them (B5)



## B5 [Solved]: Purged K-Fold vs. Shuffled K-Fold — Task

- **Question:** how much accuracy does shuffling invent when there is nothing to predict? A synthetic diagnostic, no market data
- **Inputs** (notebook section B5, seed 42): a random walk of 3,000 days; label = sign of the 20-day forward return, spans  $[i, i + 20]$ ; 5 features = 20-day means of independent noise; true accuracy 50%
- **Tasks**
  1. Audit the supplied `PurgedKFold`: find the line that purges before the test fold, the line that applies the embargo and the one that purges after it; run it on the A7 example.
  2. Fit a random forest ([Breiman \(2001\)](#); 200 trees, minimum leaf 5, seed 42) under shuffled 5-fold, contiguous 5-fold and purged 5-fold with a 1% embargo (30 observations).
  3. Report the five fold accuracies and their mean for each scheme.
- **Report:** the A7 check, a chart and a table of fold accuracies
- **Interpretation**
  - ▶ in this design, what removes the leak: purging, or not shuffling?
  - ▶ would the gap shrink with 1-day labels?

## B5 [Solved]: Solution



- On the A7 example the class removes  $\{5, 6, 7, 12, 13, 14\}$  for the test fold  $\{8, \dots, 11\}$ : the step-by-step answer

| Scheme     | Fold accuracies; mean |       |       |       |        |              |
|------------|-----------------------|-------|-------|-------|--------|--------------|
| shuffled   | 0.707                 | 0.678 | 0.698 | 0.700 | 0.687; | <b>0.694</b> |
| contiguous | 0.480                 | 0.543 | 0.553 | 0.452 | 0.472; | 0.500        |
| purged     | 0.485                 | 0.510 | 0.540 | 0.473 | 0.477; | <b>0.497</b> |

- Purged training sets keep 2,350–2,380 of 2,400 observations
- Interpretation: here contiguous folds alone remove most of the leak, because only the fold boundaries share returns; purging makes the boundaries clean. The leak needs overlapping labels *and* persistent features (Bergmeir, Hyndman & Koo (2018)); this one seed shows it for this design only

## The Bitcoin Modelling Frame (B6–B8, B12, B14)

- Target:  $y_t = 1\{\ln P_{t+5} - \ln P_t > 0\}$ , 5 calendar days (crypto trades daily); label span  $[t, t + 5]$
- Rows: every day with all features and a complete 5-day window:  $n = 4,181$ , 4 Apr 2015 – 13 Sep 2026, 54.8% “up”
- Validation: `PurgedKfold(5, t1, pct_embargo=0.01)`: 5 contiguous folds of 836–837 days, embargo 41 days; the same folds for every model

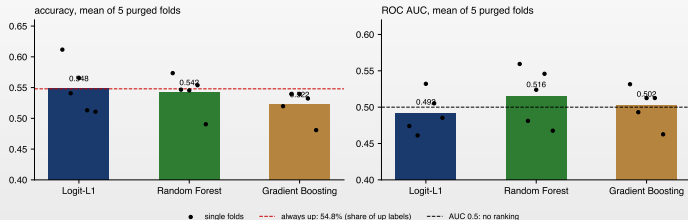
| 11 features (build_features)                                                                                  | Definition, information up to day $t$                             |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <code>mom_1</code> , <code>mom_5</code> , <code>mom_20</code> , <code>mom_60</code> ,<br><code>mom_120</code> | $x_t - x_{t-h}$                                                   |
| <code>vol_20</code> , <code>vol_60</code> , <code>vol_ratio</code>                                            | SD of $r_t$ over 20 and 60 days $\times \sqrt{365}$ ; their ratio |
| <code>dist_ma50</code> , <code>dist_ma200</code>                                                              | $x_t$ minus the mean of $x$ over the last 50 or 200 days          |
| <code>rsi_14</code>                                                                                           | relative strength index, 14 days, divided by 100                  |

- Models: Logit-L1 (standardised,  $C = 0.05$ , [Tibshirani \(1996\)](#)); RF (400 trees, minimum leaf 50, [Breiman \(2001\)](#)); GB (histogram, depth 3, learning rate 0.03, 200 trees, [Friedman \(2001\)](#)); seed 42
- Saved for later exercises: one row per date with the label, the fold, the three out-of-fold probabilities and the constant forecast (training share of “up” in that fold)

## B7a [Solved]: Three Models against “Always Up” — Task

- ▣ **Question:** does any of three models predict the 5-day direction of Bitcoin better than “always up”?
- ▣ **Data:** the modelling frame of the previous slide
- ▣ **Tasks**
  1. Fit Logit-L1, RF and GB on each purged training fold and predict the test fold; save the out-of-fold probabilities.
  2. For each model, compute the accuracy ( $\hat{p} > 0.5$ ) and the ROC AUC in each test fold, and their means (**fold means**).
  3. Compute the “always up” accuracy of each test fold, its share of up labels, and the mean over folds.
  4. Compute the **pooled** AUC: one AUC over all 4,181 out-of-fold probabilities together.
- ▣ **Report:** a table (accuracy, AUC, fold SD), the baseline, the chart with fold points, the share of “up” predictions
- ▣ **Interpretation:** why is 54.8% accuracy not evidence of skill here? Notebook: section B7

## B7a [Solved]: Solution



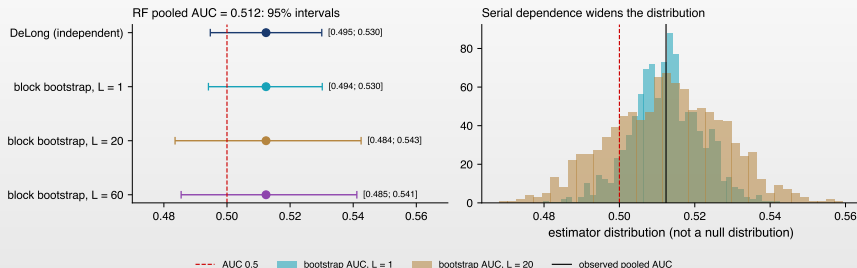
|                     | Logit-L1 | RF    | GB    |
|---------------------|----------|-------|-------|
| Accuracy, fold mean | 54.8%    | 54.2% | 52.2% |
| AUC, fold mean      | 0.492    | 0.516 | 0.502 |
| AUC, pooled         | 0.466    | 0.512 | 0.498 |
| Predicted "up"      | 99.6%    | 70.3% | 68.5% |

- "Always up": fold shares 61.3%, 53.5%, 56.6%, 51.3%, 51.3%, mean 54.8%; no model beats it; Logit-L1 is "always up"
- Fold mean and pooled AUC differ: five rankings averaged vs. one ranking over all days, which also mixes the folds' calibration
- Interpretation: 54.8% is the share of up weeks, not a skill; the next question is how uncertain the RF's 0.512 is (B7b)

## B7b [Solved]: A Dependence-Robust CI for the AUC — Task

- ▣ **Question:** is the RF's pooled AUC of 0.512 distinguishable from 0.5?
- ▣ **Data:** the saved RF out-of-fold probabilities and labels of B7a ( $n = 4,181$ );  $\text{AUC} = P(\hat{p}_+ > \hat{p}_-) + \frac{1}{2}P(\hat{p}_+ = \hat{p}_-)$ , 0.5 = no ranking
- ▣ **Tasks**
  1. Compute the DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)), the 95% CI,  $z = (\text{AUC} - 0.5)/\text{SE}$  and the two-sided  $p$ -value.
  2. Resample the rows with a circular block bootstrap (Künsch (1989)): blocks of  $L \in \{1, 20, 60\}$  consecutive days, 1,000 replications, seed 42; recompute the AUC each time.
  3. Report the bootstrap SE, the percentile 95% CI and the share of replications at or below 0.5 for each  $L$ .
- ▣ **Report:** an interval chart and the bootstrap distributions for  $L = 1$  and  $L = 20$
- ▣ **Interpretation**
  - ▶ why is the block-bootstrap interval wider?
  - ▶ which block length would you defend?
- ▣ Notebook: section B7

## B7b [Solved]: Solution



- DeLong: SE 0.0090, CI [0.495; 0.530],  $z = 1.37$ ,  $p = 0.17$ ;  $L = 1$ : SE 0.0091, CI [0.494; 0.530], as DeLong
- $L = 20$ : SE 0.0150, CI [0.484; 0.543], 20.8% of replications  $\leq 0.5$ ;  $L = 60$ : SE 0.0141, CI [0.485; 0.541]
- Interpretation: overlapping 5-day labels and volatility clusters make neighbouring days dependent; the dependence-robust SE is 1.7 times the DeLong SE. A block of at least the label horizon, stable from 20 to 60 days:  $L = 20$  is defensible. The histograms are estimator distributions, not null distributions

## B8 [Proposed]: Diebold–Mariano on Probability Forecasts — Task

- **Model:** B7a and B7b [Solved]
- **Question:** does any model forecast the probability of an up week better than a constant?
- **Inputs:** the saved probabilities of B7a, clipped to  $[10^{-6}, 1 - 10^{-6}]$ ; log loss  $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$ ; worked row, 13 Jul 2015,  $y_t = 0$ : constant  $\hat{p} = 0.529$ ,  $\ell = -\ln(1 - 0.529) = 0.753$ ; RF  $\hat{p} = 0.496$ ,  $\ell = 0.684$
- **Tasks**
  1. Compute the mean log loss of Logit-L1, RF, GB and the constant.
  2. For RF vs. Logit-L1, RF vs. constant and GB vs. constant, form  $d_t = \ell_t^A - \ell_t^B$  ( $d_t > 0$ : model A worse) and regress  $d_t$  on a constant with Newey–West errors ([Newey & West \(1987\)](#)), lags 4 and 20: the DM statistic is the  $t$ -value ([Diebold & Mariano \(1995\)](#)), two-sided.
  3. Compute the naive i.i.d.  $t$  of  $\bar{d}$  and compare.
- **Report:** a table of log losses,  $\bar{d}$ , DM (lags 4, 20) with  $p$ -values, the naive  $t$ ; a chart of cumulative  $d_t$
- **Interpretation:** can a model with AUC above 0.5 lose to a constant in log loss? Pooled CV probabilities are descriptive: a deployment claim needs walk-forward forecasts. Notebook: section B8

## C2 [Proposed]: Critique an AI Answer — the Answer

□ **Prompt** to an AI assistant: “Predict tomorrow’s direction of Bitcoin from past returns and evaluate the model out of sample with 5-fold cross-validation”

□ **Answer** (btc: daily closes 2014-09-17 to 2026-09-18; imports omitted):

```
1  r = np.log(btc).diff()
2  X = pd.DataFrame({'r1': r, 'r5': r.rolling(5).sum(), 'vol20': r.rolling(20).std(),
3                  'trend10': r.rolling(10, center=True).mean()})
4  data = pd.concat([X, r.shift(-1).rename('fwd')], axis=1).dropna()
5  y = (data['fwd'] > 0).astype(int) # 1 if tomorrow is up
6  Xs = StandardScaler().fit_transform(data[X.columns])
7  cv = KFold(n_splits=5, shuffle=True, random_state=0)
8  model = LogisticRegression(penalty='l1', C=0.1, solver='liblinear')
9  auc = cross_val_score(model, Xs, y, cv=cv, scoring='roc_auc')
10 print(round(auc.mean(), 3))
```

□ “Output: 0.697. An out-of-sample AUC of 0.70 on 4,361 days is a strong, tradable signal; cross-validation is out of sample, so there is no overfitting”

## C2 [Proposed]: Critique an AI Answer — Task

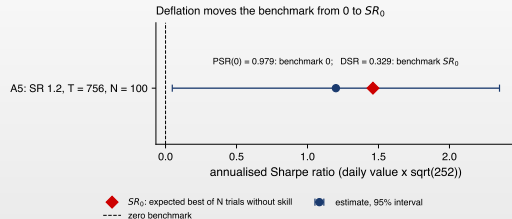
- ▣ **Model:** B5 and B7a [Solved]; B6 (extension) is the same trap in a larger model
- ▣ **Tasks**
  1. Give five annotated findings, each with a line number: two leaks of test information, one validation design that does not match deployment, two unsupported claims.
  2. For each, say how you would detect it: a check on the output, a result of B5 or B7a, a rule of the primer.
  3. Write one corrected pipeline: trailing features, the scaler inside each training fold (`make_pipeline`), `TimeSeriesSplit(n_splits=5, gap=20)`, mean AUC over the 5 test folds.
  4. Report a before/after table: the AI code, then each correction added one at a time.
  5. Log the errors in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.
- ▣ **Interpretation**
  - ▶ which days enter `trend10` on day  $t$ ?
  - ▶ which of them is the target?
- ▣ An unusually high AUC is a reason to inspect the code, not a proof of leakage

## A5 [Solved]: PSR and DSR, a Worksheet — Task

- **Question:** is an annualised Sharpe ratio of 1.2 on three years convincing if it is the best of 100 trials?
- **Inputs** (equity strategy,  $q = 252$ ):  $\widehat{SR}_{ann} = 1.2$ ,  $T = 756$  daily returns,  $\hat{\gamma}_3 = -0.4$ , ordinary kurtosis  $\hat{\gamma}_4 = 5$ ;  $N = 100$  independent trials,  $V = \text{Var}[\widehat{SR}_n] = 1/T$ ; Euler's constant  $\gamma_E = 0.5772$  (Bailey & López de Prado (2012); Bailey & López de Prado (2014))
- **Tasks**
  1. Daily  $\widehat{SR} = 1.2/\sqrt{252}$  and the denominator  $D = \sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}$ .
  2.  $z = \widehat{SR} \sqrt{T-1}/D$  and  $PSR(0) = \Phi(z)$ .
  3.  $SR_0 = \sqrt{V} [(1 - \gamma_E) \Phi^{-1}(1 - \frac{1}{N}) + \gamma_E \Phi^{-1}(1 - \frac{1}{N_e})]$ , daily and annualised.
  4.  $DSR = \Phi((\widehat{SR} - SR_0) \sqrt{T-1}/D)$ .
- **Report:** every intermediate number, PSR(0) and DSR; the derivation of  $D$  is in the extension (A5 Extended)
- **Interpretation**
  - ▶ under i.i.d. returns, a zero benchmark and 100 trials, what does the DSR say?
  - ▶ which  $N$  would change it?
- Notebook: section A5

## A5 [Solved]: Solution

1.  $\widehat{SR} = 1.2/\sqrt{252} = 0.07559$ ;  
 $D = \sqrt{1 + 0.4 \cdot 0.07559 + 1 \cdot 0.07559^2} = 1.01782$
  2.  $z = 0.07559 \cdot \sqrt{755}/1.01782 = 2.041$ ;  
 $PSR(0) = \Phi(2.041) = 0.979$
  3.  $\sqrt{V} = 1/\sqrt{756} = 0.03637$ ;  $\Phi^{-1}(0.99) = 2.326$ ,  
 $\Phi^{-1}(1 - \frac{1}{100e}) = 2.680$ ;  
 $SR_0 = 0.03637 [0.4228 \cdot 2.326 + 0.5772 \cdot 2.680] = 0.0920$  a  
 day, 1.46 annualised
  4.  $z = (0.07559 - 0.0920)\sqrt{755}/1.01782 = -0.444$ ;  
 $DSR = \Phi(-0.444) = 0.329$
- Interpretation: luck alone gives a best annualised SR of 1.46 among 100 i.i.d. zero-skill trials, above the observed 1.2; the evidence does not clear the illustrative 0.95 threshold. With  $N = 1$  the DSR equals  $PSR(0) = 0.979$



## Team Project: Choose One Route

- ▣ **Standard:** one asset of your choice through B1, B3, B7a, B7b and B10, with the candidate-return matrix of every rule tried saved
- ▣ **Advanced:** C1 (foundation model vs. boosting) or C3 (a no-arbitrage SDF on deciles), in the extension packs
- ▣ Minimum evidence: code as a Quantlet, one chart, the test against a benchmark, the number of trials; AI use in `AI_USE.md`, caught errors in `AI_ERRORS.md`
- ▣ A negative result, honestly documented, is a good result

## After the Lecture: Two Solved Exercises to Redo

- Do these two after [Chapter 13](#): they use estimators and arguments that the lecture develops
  - ▶ B1: the exact local Whittle estimator ([Shimotsu & Phillips \(2005\)](#)) and the inference on the memory  $\delta$  of the Bitcoin log price; model: A1
  - ▶ B10: a grid of 120 correlated trading rules, the eigenvalues of their correlation matrix and the effective number of trials in the DSR; model: A5
- Both are [Solved]: redo them in the notebook (sections B1 and B10) and compare with the slides

## B1 [Solved]: Memory of the Bitcoin Log Price — Task

### ▣ Questions

- ▶ which  $d$  does the minimum-ADF rule choose for Bitcoin?
- ▶ what is the memory  $\delta$  of the log price?

- ▣ **Data:**  $x_t = \ln P_t$ , BTC 2014-09-17 to 2026-09-18 ( $n = 4,385$ ); supplied: `frac_diff_ffd`, `exact_local_whittle`, `local_whittle`

### ▣ Tasks

1. For  $d \in \{0, 0.05, \dots, 1\}$ , filter  $x_t$  ( $\tau = 10^{-4}$ ), run `adfuller(x, maxlag=1, regression='c', autolag=None)` (Dickey & Fuller (1979)) and record the statistic and  $\text{Corr}(\tilde{x}_t, x_t)$  on the dates the filter keeps.
2. Report  $d^*$ , the smallest  $d$  with a statistic below the 5% critical value.
3. Estimate  $\delta$  with 95% CIs ( $m = \lfloor n^{0.65} \rfloor$  of each series): ELW (Shimotsu & Phillips (2005)) for  $x_t$  and the FFD series at  $d^*$ , LW (Robinson (1995)) for  $r_t$  and  $|r_t|$ .
4. Test  $H_0 : \delta = 1$  for  $x_t$  with  $z = (\hat{\delta} - 1)/\text{SE}$ , from the unrounded estimate.

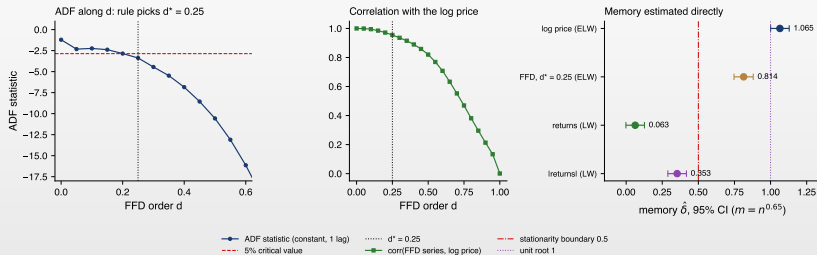
- ▣ **Report:** the ADF–correlation chart,  $d^*$ , a table of the four estimates with CIs,  $z$  and  $p$

### ▣ Interpretation

- ▶ is the FFD series at  $d^*$  stationary?
- ▶ where does the memory of Bitcoin sit?

- ▣ Notebook: section B1

## B1 [Solved]: Solution

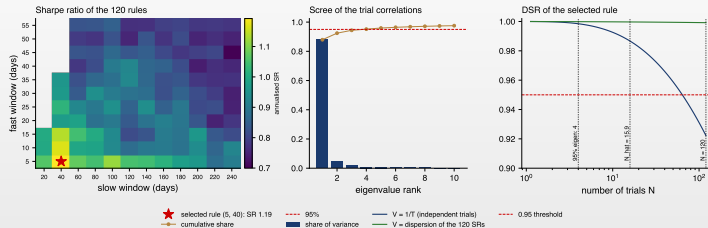


- ADF:  $-1.21$  ( $d = 0$ ),  $-2.32$  (0.05),  $-2.24$  (0.10),  $-2.39$  (0.15),  $-2.85$  (0.20),  $-3.38$  (0.25); 5% value  $-2.86$ :  $d^* = 0.25$ , correlation 0.955; the path is not monotone in  $d$
- ELW log price 1.065 [1.000; 1.129] ( $m = 232$ ); FFD at  $d^*$  0.814 [0.747; 0.880]; LW returns 0.063 [ $-0.002$ ; 0.127];  $|r_t|$  0.353 [0.289; 0.417]
- $z = (1.0647 - 1)/0.0328 = 1.97$ , two-sided  $p = 0.049$ : a borderline rejection towards  $\delta > 1$  (bubbles), never towards stationarity
- Interpretation: the FFD series has  $\hat{\delta} \approx 1 - d$  over the band and its CI lies far above  $\frac{1}{2}$ : not stationary; Bitcoin's long memory is in volatility  $|r_t|$

## B10 [Solved]: Grid Search, DSR and the Effective Number of Trials — Task

- ▣ **Question:** after searching 120 moving-average rules on Bitcoin, is the best one better than luck?
- ▣ **Rules:** fast window  $f \in \{5, 10, \dots, 55\}$ , slow  $s \in \{20, 40, \dots, 240\}$ ,  $s > f$ :  $N = 120$ ; signal on day  $t$ :  $1\{\bar{x}_{f,t} > \bar{x}_{s,t}\}$  (means of the log price), held on day  $t + 1$ :  $r_{t+1}^{(f,s)} = \text{signal}_t \cdot r_{t+1}$ ; long/cash, gross of costs, risk-free rate 0
- ▣ **Dates:** common start on the first day every rule has a signal, 15 May 2015, to 18 Sep 2026:  $T = 4,145$  days,  $q = 365$
- ▣ **Tasks**
  1. Build the  $T \times 120$  return matrix; pick the rule with the best SR; report its SR, skewness, kurtosis and PSR(0) (Bailey & López de Prado (2012)).
  2. From the  $120 \times 120$  correlation matrix: eigenvalues, the number reaching 95% of the variance, and  $\hat{N} = \bar{\rho} + (1 - \bar{\rho})N$  with  $\bar{\rho}$  the mean off-diagonal correlation (heuristics, not theorems).
  3. Compute  $SR_0$  and the DSR (Bailey & López de Prado (2014)) for  $N = 120$ ,  $\hat{N}$  and the 95% count, with  $V = 1/T$ .
- ▣ **Report:** a Sharpe heatmap with the selected rule, the scree plot, a DSR table
- ▣ **Interpretation:** which  $N$  and which  $V$  would you report to an investment committee? Notebook: section B10

## B10 [Solved]: Solution



| $V = 1/T$     | $N = 120$ | $\hat{N} = 15.9$ | $N = 4$ |
|---------------|-----------|------------------|---------|
| $SR_0$ (ann.) | 0.77      | 0.53             | 0.31    |
| DSR           | 0.922     | 0.987            | 0.998   |

- Best rule (5, 40): SR 1.19 annualised (buy-and-hold 0.78); skewness 0.13, kurtosis 12.8;  $PSR(0) = 1.000$
- $\lambda_1 = 105.5$  explains 87.9%; 4 eigenvalues reach 95%;  $\bar{\rho} = 0.875$ ,  $\hat{N} = 15.9$
- Interpretation: counting 120 correlated trials as independent over-deflates; the DSR tests “above luck”, not “above buy-and-hold” (B13), and the winner was chosen on the full sample: a claim needs a frozen rule and an untouched holdout (lecture, Bitcoin grid)

## Extension Packs: What Each One Needs

| Pack                            | Exercises                                       | Prerequisites                                                |
|---------------------------------|-------------------------------------------------|--------------------------------------------------------------|
| 1. Memory                       | A2, B2                                          | A1, B1                                                       |
| 2. Labels                       | A3 Extended [Solved], A4, B4                    | A3, B3                                                       |
| 3. Validation and leaks         | A8, B6                                          | A7, B5, the modelling frame                                  |
| 4. Predictive inference         | A9 [Solved], A10, B7c [Solved], B12 [Solved]    | B7a, B7b, B8                                                 |
| 5. Strategies and selection     | A5 Extended [Solved], A6, B9 [Solved], B11, B13 | A5, B10                                                      |
| 6. Inference with many controls | B14                                             | the lecture example (VIX)                                    |
| 7. Projects                     | C1, C3                                          | <a href="#">Chapter 14</a> (C1); the lecture case study (C3) |

- Each exercise has its notebook section with the same label; do them in order within a pack
- C1 and C3 are possible starts for the team project

## A2 [Proposed]: Window Length and ADF Over-Rejection — Task

- ▣ **Model:** A1 [Solved]
- ▣ **Question:** how often does ADF reject a unit root in the FFD series of a pure random walk?
- ▣ **Simulation** (`rng = default_rng(42)`): for  $d \in \{0.3, 0.5\}$  and each of 500 paths, compute  $K$ , draw  $T + K$  innovations  $\varepsilon_s \sim N(0, 1)$  with  $T = 2,500$ , set  $X_t = \sum_{s \leq t} \varepsilon_s$  ( $X$  starts at the first draw), apply the finite filter and keep exactly  $T$  values; then repeat with  $d = 0.25$  and with no filter (control)
- ▣ **Tasks**
  1. Run `adfuller(f, maxlag=1, regression='c', autolag=None)` on each path and record rejections at 5%.
  2. On the same paths, run the test with lags chosen by AIC (`autolag='AIC'`).
  3. Report each rejection frequency with its Monte Carlo SE  $\sqrt{\hat{p}(1 - \hat{p})/500}$ .
- ▣ **Interpretation:** every path is  $I(1)$ 
  - ▶ why does a one-lag ADF confuse a scaled random walk with a stationary series?
  - ▶ which  $d$  would make an  $I(1)$  price stationary?
- ▣ **Notebook:** section A2

## B2 [Proposed]: Memory of Ether, a Trap — Task

- ▣ **Model:** B1 [Solved]
- ▣ **Question:** how robust is the minimum-ADF rule on an asset that grew about 1,000-fold?
- ▣ **Data:** ETH-USD log closes; three specifications, each with a full  $d$ -grid scan  $\{0, 0.05, \dots, 1\}$  and one ADF lag

| Row | Sample                                  | Deterministic terms | 5% critical value |
|-----|-----------------------------------------|---------------------|-------------------|
| 1   | 2015-08-07 – 2026-09-18 ( $n = 4,061$ ) | constant            | −2.86             |
| 2   | 2015-08-07 – 2026-09-18                 | constant + trend    | −3.41             |
| 3   | 2018-01-01 – 2026-09-18 ( $n = 3,183$ ) | constant            | −2.86             |

- ▣ **Tasks:** (1) report  $d^*$  and the ADF values at  $d = 0$  and  $0.05$  for each row; (2) ELW with 95% CI for the log price on rows 1 and 3; (3) LW for  $|r_t|$  on the full sample
- ▣ **Interpretation**
  - ▶ which  $d$  would you use in a model?
  - ▶ why that  $d$ ?
- ▣ **Notebook:** section B2

## A3 Extended [Solved]: Hitting Probabilities — Task

- ▣ **Question:** what share of events should end at the vertical barrier if prices were a Brownian motion?
- ▣ **Setting:** log price  $\sigma W_t$ ,  $W_t$  a standard Brownian motion (no drift), barriers  $+a = pt \sigma \sqrt{h}$  and  $-b = -sl \sigma \sqrt{h}$ ;  $\tau =$  first time either is hit; martingales  $W_t$  and  $W_t^2 - t$ ; stopping time  $\tau$  with  $\mathbb{E}[\tau] < \infty$  (Karatzas & Shreve (1998))
- ▣ **Tasks**
  1. Without a vertical barrier, derive  $P(\text{upper first})$  and  $\mathbb{E}[\tau]$  by optional stopping.
  2. For  $pt = sl = 1$  and  $h = 10$ , compute  $P(\tau > h)$  under continuous monitoring with
 
$$P(\tau > h) = \frac{4}{\pi} \sum_{n \geq 0} \frac{(-1)^n}{2n+1} e^{-(2n+1)^2 \pi^2 / 8}.$$
  3. Approximate daily monitoring by moving each barrier out by  $0.5826 \sigma$  (Broadie, Glasserman & Kou (1997)); then simulate 400,000 paths of 10 Gaussian daily steps (seed 42) and compare.
- ▣ **Interpretation:** B3 finds 59.0% timeouts on Bitcoin: which part of the gap to the Brownian value is monitoring, which is the data?

## A3 Extended [Solved]: Solution

1.  $W_t$  is a martingale:  $0 = p a - (1 - p) b \Rightarrow p = \frac{b}{a+b} = \frac{sl}{pt+sl}$ ;  $\sigma^2 W_t^2 - \sigma^2 t$  is a martingale:  
 $\sigma^2 \mathbb{E}[\tau] = p a^2 + (1 - p) b^2 = ab \Rightarrow \mathbb{E}[\tau] = pt \cdot sl \cdot h \text{ days}$
  2. Continuous monitoring, exact:  $P(\tau > h) = 1.2732 \cdot 0.2912 - \dots = 0.371$
  3. Daily monitoring, approximate:  $c = 1 + 0.5826/\sqrt{10} = 1.184$ , replacing  $\pi^2/8$  by  $\pi^2/(8c^2)$  gives 0.528;  
simulated with daily steps: 0.517 (SE 0.001)
- Interpretation: discrete monitoring explains most of the gap from 37% to 59%; the remaining 7 points may come from fat tails, volatility clustering and error in the EWMA volatility, which this comparison does not separate

## A4 [Proposed]: Asymmetric Barriers — Task

- ▣ **Model:** A3 and A3 Extended [Solved]
- ▣ **Inputs:**  $P_{t_0} = 50$ ,  $\sigma_{t_0} = 2\%$ ,  $h = 4$ ,  $pt = 2$ ,  $sl = 1$ ; paths (i) 50, 51.5, 53.0, 49.5, 47.9 and (ii) 50, 52.0, 54.5, 53.0, 51.0
- ▣ **Tasks**
  1. Compute both barriers, draw them on the two paths and label each path.
  2. Optional (drift): for  $X_t = \mu t + \sigma W_t$  without a vertical barrier,  $s(x) = e^{-2\mu x/\sigma^2}$  solves  $\frac{\sigma^2}{2}s'' + \mu s' = 0$ , so  $s(X_t)$  is a martingale; derive  $P(\text{upper first}) = \frac{e^{2\mu b/\sigma^2} - 1}{e^{2\mu b/\sigma^2} - e^{-2\mu a/\sigma^2}}$  and evaluate it for  $\sigma = 3\%$ ,  $\mu = 0.1\%$  a day,  $h = 10$ .
- ▣ **Report:** the barriers, the labels, the formula and its value
- ▣ **Interpretation:** the formula has no vertical barrier: which paths would a vertical barrier at  $h$  censor? Notebook: section A4

## B4 [Proposed]: Asymmetric Barriers on Bitcoin — Task

- ▣ **Model:** B3 [Solved] and A3 Extended
- ▣ **Data:** the events, EWMA volatility and labelling of B3,  $h = 10$ ; settings  $(pt, sl) = (2, 1)$  and  $(1, 2)$
- ▣ **Tasks**
  1. Report the exit shares, the share of +1 labels and the average holding period for both settings.
  2. Compute the upper share among horizontal exits,  $\text{upper}/(\text{upper} + \text{lower})$ .
  3. Compare it with  $sl/(pt + sl)$  (no vertical barrier) and with a simulation: 400,000 paths of 10 Gaussian daily steps in units of  $\sigma$ , barriers at  $pt\sqrt{10}$  and  $-sl\sqrt{10}$ , seed 42, zero drift and the BTC drift  $\bar{r}/\hat{\sigma} = 0.034$  a day; give the simulation SE.
- ▣ **Interpretation:** how does the choice change the class balance a classifier sees? The asymmetry is a modelling hypothesis, not an optimal trading rule. Notebook: section B4

## A8 [Proposed]: Purging with Shorter Labels — Task

- ▣ **Model:** A7 [Solved]
- ▣ **Inputs:** as A7, with closed label spans  $[i, \min(i + 2, 19)]$  ( $h = 2$ ), embargo  $e = 1$ , test fold  $\{12, \dots, 15\}$
- ▣ **Tasks**
  1. On a timeline of indices 0–19, fill a table with the columns purged before, embargo, purged after and union.
  2. Check the count with the A7 formula.
  3. Interior fold with  $n = 250$ ,  $K = 5$  ( $L = 50$ ),  $h = 20$ ,  $e = 1$ : compute the share lost; repeat with  $K = 10$  ( $L = 25$ ).
- ▣ **Interpretation**
  - ▶ what does purging cost when labels are long and folds are short?
  - ▶ how would you respond?
- ▣ **Notebook:** section A8

## B6 [Proposed]: Find the Leak — Task

- ▣ **Model:** B5 [Solved] and C2
- ▣ **Data:** the modelling frame, plus a 12th feature built in the notebook as `ma11 = r.rolling(11, center=True).mean()`; RF with 300 trees (not 400 as in B7a), minimum leaf 50, seed 42, purged 5-fold; 4,181 rows
- ▣ **Tasks**
  1. Write the dates  $t - 5, \dots, t + 5$  that enter `ma11` on day  $t$  and mark those inside the target window  $t + 1, \dots, t + 5$ ; rank the 12 features by MDI.
  2. Compute the correlation of `ma11` and of the trailing `r.rolling(11).mean()` with the 5-day forward return.
  3. Report accuracy and AUC for: 11 features + centred `ma11`; 11 + trailing; the 11 base features, same rows, folds and settings.
- ▣ **Interpretation:** why can purging not catch this leak? Give a second example of look-ahead in a feature pipeline.  
Notebook: section B6

## A9 [Solved]: AUC Step by Step — Task

### ▣ Questions

- ▶ what does an AUC measure?
- ▶ how uncertain is it with six observations?

- ▣ **Inputs:** scores of 3 positives (0.80, 0.55, 0.40) and 3 negatives (0.60, 0.35, 0.20);  $U$  = number of (positive, negative) pairs ranked correctly,  $A = U/9$ ; Hanley–McNeil SE ([Hanley & McNeil \(1982\)](#)):

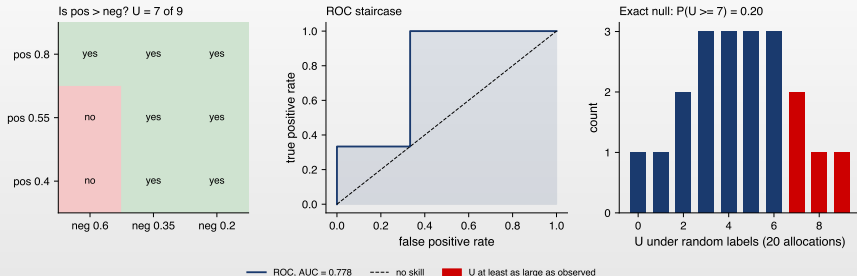
$$SE^2 = \frac{A(1-A) + (n_1-1)(Q_1-A^2) + (n_0-1)(Q_2-A^2)}{n_1 n_0}, \quad Q_1 = \frac{A}{2-A}, \quad Q_2 = \frac{2A^2}{1+A}$$

### ▣ Tasks

1. Fill the  $3 \times 3$  comparison grid (is the positive above the negative?) and compute  $U$  and  $A$ .
2. Draw the ROC staircase and check that its area equals  $A$ .
3. Compute the SE and the Wald  $z = (A - 0.5)/SE$ , two-sided.
4. Enumerate the 20 ways to label 3 of the 6 scores as positive, compute  $U$  for each and the exact  $P(U \geq 7)$ .

- ▣ **Interpretation:** which assumption of the SE fails for overlapping 5-day labels? Notebook: section A9

## A9 [Solved]: Solution



- 0.80 beats 3 negatives, 0.55 beats 2, 0.40 beats 2:  $U = 7$ ,  $A = 0.778$ ; (2) the staircase has area  $\frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot 1 = \frac{7}{9}$
- $Q_1 = 0.636$ ,  $Q_2 = 0.681$ ,  $SE = 0.207$ ,  $z = 1.34$ ,  $p = 0.18$  (asymptotic, unreliable with six points)
- Exact:  $P(U \geq 7) = 4/20 = 0.20$ , two-sided  $p = 0.40$  under exchangeable labels: six observations cannot separate skill from luck
- ☐ Both SEs assume independent observations; overlapping labels share returns, so the effective sample is smaller: B7b uses a block bootstrap instead

## A10 [Proposed]: AUC as a U-Statistic and the DeLong Variance — Task

▣ **Model:** A9 [Solved]

▣ **Inputs:** 4 positives (0.70, 0.62, 0.48, 0.45), 4 negatives (0.66, 0.50, 0.40, 0.30); kernel  $\psi(s_i, s_j) = 1\{s_i > s_j\} + \frac{1}{2}1\{s_i = s_j\}$

▣ **Tasks**

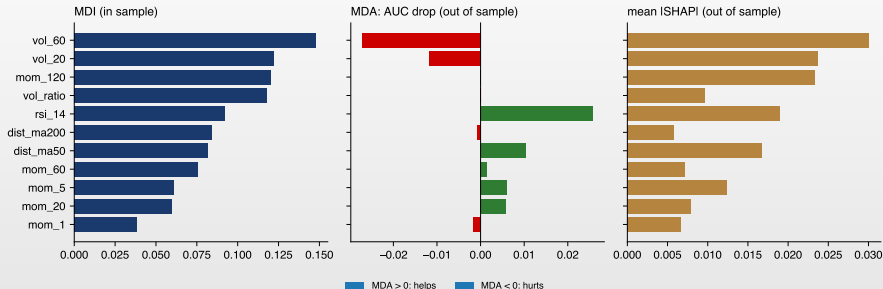
1. Fill the  $4 \times 4$  grid of  $\psi$ ; compute  $U$ , the AUC and its Hanley–McNeil SE.
2. Row means  $V_{10}(i)$  and column means  $V_{01}(j)$  of the grid (placement values); their sample variances  $S_{10}$ ,  $S_{01}$  with denominator  $n - 1$ .
3. The DeLong SE  $= \sqrt{S_{10}/n_1 + S_{01}/n_0}$  (DeLong, DeLong & Clarke-Pearson (1988)).
4. Optional proof: by the Hoeffding decomposition the AUC is asymptotically the sum of two independent averages of placement values.

▣ **Interpretation:** the argument is asymptotic, not an exact small-sample guarantee: what does that mean with 8 scores? Notebook: section A10

## B7c [Solved]: Do Importance Measures Agree? — Task

- ▣ **Question:** do in-sample and out-of-sample importance measures rank the 11 features the same way?
- ▣ **Data:** the modelling frame, with a separate chronological split: RF (400 trees, minimum leaf 50) fitted on the first 70% (to 8 Apr 2023), evaluated after a gap of 5 + 41 days, from 24 May 2023; a single split, because importances need one fitted model
- ▣ **Tasks**
  1. MDI from the fitted forest; MDA as the drop in test AUC when a feature is permuted (10 repeats); mean |SHAP| on 300 test days ([Lundberg & Lee \(2017\)](#)).
  2. Spearman rank correlations between the three measures.
- ▣ **Interpretation**
  - ▶ what does a negative MDA mean?
  - ▶ why do correlated features blur all three measures?
- ▣ Notebook: section B7

## B7c [Solved]: Solution



- MDI ranks vol\_60, vol\_20, mom\_120 first; MDA is negative for both volatility features and largest for rsi\_14 (0.026)
- Spearman: MDI-SHAP 0.75, MDI-MDA -0.42, MDA-SHAP -0.10: in-sample and out-of-sample rankings disagree
- Interpretation: a negative MDA means the permuted feature scores better than the real one on new data: a sign of overfitting, within sampling noise; correlated features share and dilute importance, and permutation extrapolates off the data

## B12 [Solved]: Pesaran–Timmermann and Clark–West — Task

### ▣ Questions

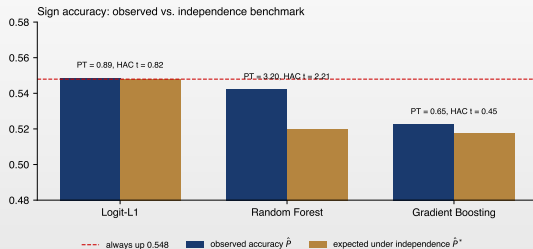
- ▶ do the predicted signs carry information?
- ▶ do the features add information to a constant forecast?

- ▣ **Data:** the saved probabilities of B7a and the constant of B8;  $x_t = 1\{\hat{p}_t > 0.5\}$ . Caveat before you start: Clark–West (Clark & West (2007)) is derived for nested linear models and genuine out-of-sample forecasts; on penalised-logit probabilities from pooled purged CV it is an approximate, descriptive adjustment

### ▣ Tasks

1. For each model:  $\hat{P}$  = share of correct signs,  $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$  and  $PT = (\hat{P} - \hat{P}^*) / \sqrt{\hat{V}(\hat{P}) - \hat{V}(\hat{P}^*)}$  (Pesaran & Timmermann (1992); the variances are in the lecture Appendix).
  2. The slope  $t$  of  $y_t = a + b x_t + u_t$  with Newey–West errors (4 lags), two-sided.
  3. Logit-L1 vs. constant:  $f_t = (y_t - \hat{p}_{0,t})^2 - [(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2]$ ; CW = HAC  $t$  of  $\bar{f}$ , one-sided (positive = the features help); DM on the Brier loss, two-sided.
- ▣ **Interpretation:** why can DM and Clark–West disagree for nested models? Notebook: section B12

## B12 [Solved]: Solution



|          | Pred. up | $\hat{P}$ | $\hat{P}^*$ | PT   | $t_{HAC}$ |
|----------|----------|-----------|-------------|------|-----------|
| Logit-L1 | 99.6%    | 0.548     | 0.548       | 0.89 | 0.82      |
| RF       | 70.3%    | 0.542     | 0.519       | 3.20 | 2.21      |
| GB       | 68.5%    | 0.522     | 0.518       | 0.65 | 0.45      |

- RF: accuracy 54.2% is below “always up” (54.8%), yet its signs are not independent of the outcome:  $PT = 3.20$ ; with HAC errors  $t = 2.21$ ,  $p = 0.03$
- Clark–West, Logit-L1 vs. constant:  $t = -0.56$ , one-sided  $p = 0.71$ ; DM on Brier:  $t = 1.37$  (positive = logit worse),  $p = 0.17$
- DM asks whether the *estimated* logit forecasts better, and its estimation noise makes it lose under the nested null; CW adds back  $(\hat{p}_0 - \hat{p}_1)^2$  and asks whether the population coefficients are zero; here both find no signal
- Sign dependence (PT) is not the same as beating a majority baseline, nor as a better log loss (B8)

## A5 Extended [Solved]: The Variance of the Sharpe Ratio — Task

- ▣ **Question:** where does the denominator  $D$  of A5 come from?
- ▣ **Setting:**  $T$  i.i.d. returns with mean  $\mu$ , variance  $\sigma^2$ , skewness  $\gamma_3$ , ordinary kurtosis  $\gamma_4$ ;  
 $SR = \mu/\sigma = g(m_1, m_2) = m_1/\sqrt{m_2 - m_1^2}$  with  $m_j = \mathbb{E}[r^j]$
- ▣ **Tasks**
  1. Compute the gradient  $\nabla g$ .
  2. Compute  $\text{Var}(r)$ ,  $\text{Cov}(r, r^2)$  and  $\text{Var}(r^2)$  in terms of  $\mu, \sigma, \gamma_3, \gamma_4$ .
  3. Show that  $T \text{Var}(\widehat{SR}) \approx \nabla g^\top \Sigma \nabla g = 1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2$  (Lo (2002)).
- ▣ **Interpretation:** which moment makes a strategy with negative skewness look more precise than it is?

## A5 Extended [Solved]: Solution

1.  $\nabla g = \left( \frac{1}{\sigma} + \frac{\mu^2}{\sigma^3}, -\frac{\mu}{2\sigma^3} \right)$
  2.  $\text{Var}(r) = \sigma^2$ ,  $\text{Cov}(r, r^2) = \sigma^3\gamma_3 + 2\mu\sigma^2$ ,  $\text{Var}(r^2) = \sigma^4(\gamma_4 - 1) + 4\mu\sigma^3\gamma_3 + 4\mu^2\sigma^2$
  3. In  $\nabla g^\top \Sigma \nabla g$  the terms in  $\mu^2/\sigma^2$  cancel and leave  $1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2$ ; for the Normal distribution ( $\gamma_3 = 0$ ,  $\gamma_4 = 3$ )  $1 + \frac{1}{2} SR^2$
- Interpretation: with  $SR > 0$ , negative skewness adds  $|\gamma_3|SR$  to the variance and fat tails add  $\frac{\gamma_4 - 3}{4} SR^2$ : ignoring them understates the SE; A5 used  $\gamma_3 = -0.4$ ,  $\gamma_4 = 5$

## A6 [Proposed]: DSR on a Shorter Sample and the Gumbel Constant — Task

▣ **Model:** A5 and A5 Extended [Solved]

▣ **Inputs:**  $\widehat{SR}_{ann} = 1.5$ ,  $T = 504$  daily returns ( $q = 252$ ),  $\hat{\gamma}_3 = -0.2$ ,  $\hat{\gamma}_4 = 6$ ,  $N = 20$  independent trials,  $V = 1/T$ ; Euler's constant  $\gamma_E = 0.5772$  (not the skewness)

▣ **Tasks**

1. Fill the A5 worksheet:  $\widehat{SR}$ ,  $D$ ,  $PSR(0)$ ,  $SR_0$  (daily and annualised), DSR.
2. Optional: with location  $b_N = \Phi^{-1}(1 - \frac{1}{N})$  and scale  $a_N = \Phi^{-1}(1 - \frac{1}{Ne}) - b_N$ ,  $\max_n Z_n$  is approximately Gumbel; show that its mean is  $(1 - \gamma_E)b_N + \gamma_E\Phi^{-1}(1 - \frac{1}{Ne})$ .
3. Optional: check it by simulation for  $N = 100$ , 20,000 draws, seed 42, and plot the distribution.

▣ **Interpretation**

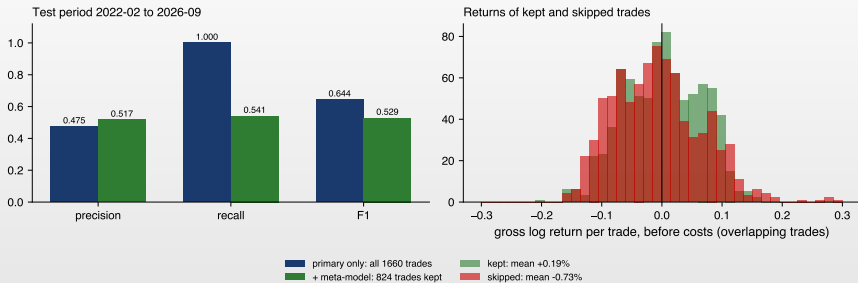
- ▶ is the evidence stronger than in A5?
- ▶ does it clear 0.95?

▣ Notebook: section A6

## B9 [Solved]: Meta-Labelling a Moving-Average Rule — Task

- ▣ **Question:** can a classifier tell when to trust a simple trend rule?
- ▣ **Primary side:**  $s_t = \text{sign}(MA_{20}(P)_t - MA_{50}(P)_t)$  on prices, one candidate trade on every eligible day (a standing daily side, not only crossings); exit by the triple barrier  $pt = sl = 1, h = 10$ ; gross meta-label  $1\{s_t \cdot r_{t_0, t_1} > 0\}$  (López de Prado (2018))
- ▣ **Secondary model:** RF (400 trees, minimum leaf 50, balanced class weights) on the 11 features plus  $s_t$ ; train on the first 60% of 4,174 events (to 9 Feb 2022), gap 10 days, test 20 Feb 2022 – 6 Sep 2026 (1,660 trades)
- ▣ **Tasks:** (1) the confusion matrix of “take if  $\hat{p} > 0.5$ ”; (2) precision, recall, F1 before (take every trade) and after; (3) the gross return per trade, kept vs. skipped
- ▣ **Interpretation:** trades overlap: this is not a portfolio backtest; what would a capital rule and costs add?  
Notebook: section B9

## B9 [Solved]: Solution



1. Taken: 426 winners, 398 losers; skipped: 362 winners, 474 losers (824 of 1,660 kept, 49.6%)
2. Before: precision 0.475, recall 1 (every winner is taken), F1 0.644; after: 0.517, 0.541, 0.529
3. Gross log return per trade: all  $-0.27\%$ , kept  $+0.19\%$ , skipped  $-0.73\%$

□ Interpretation: filtering raises precision and the return per trade while F1 falls; the next steps are a block-bootstrap test of that return, costs, a capital rule for overlapping trades and a DSR over all meta-model settings tried

## B11 [Proposed]: Probability of Backtest Overfitting — Task

- ▣ **Model:** B10 [Solved]; the lecture slide on PBO
- ▣ **Data:** the  $4,145 \times 120$  return matrix of B10, split into  $S = 16$  contiguous blocks with `np.array_split` (one block of 260 days, fifteen of 259) (Bailey, Borwein, López de Prado & Zhu (2017))
- ▣ **Warm-up:** do the steps below by hand for 4 strategies and 4 blocks
- ▣ **Tasks**
  1. For each of the  $\binom{16}{8} = 12,870$  choices of 8 in-sample blocks, pick the rule with the best in-sample SR.
  2. Rank all 120 rules out of sample in ascending order (average ranks for ties),  $\omega = \text{rank}/(N + 1)$ ,  $\lambda = \ln \frac{\omega}{1-\omega}$ .
  3. Report  $\text{PBO} = P(\lambda \leq 0)$ , the median  $\lambda$ , the median IS and OOS SR of the winner, the share with OOS SR  $< 0$ , and the histogram of  $\lambda$ .
- ▣ **Interpretation:** PBO is the share of selected winners at or below the OOS median, not the probability of losing money; is the grid overfit? Notebook: section B11

## B13 [Proposed]: Reality Check, SPA and StepM on the 120 Rules — Task

- **Model:** B10 [Solved]; supplied: `stationary_bootstrap_index` and `snooping_tests` (notebook section B13)
- **Null:**  $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$  with  $d_{k,t} = r_{k,t} - r_{0,t}$ ; benchmarks cash ( $r_0 = 0$ ) and buy-and-hold ( $r_0 = r_t$ ); these tests compare *mean* returns, the DSR concerns Sharpe ratios
- **Bootstrap** (Politis & Romano (1994)): one index sequence per replication, applied to every column (whole days), mean block 20 days ( $q = 1/20$ ), 2,000 replications, seed 42;  $\hat{\omega}_k = \sqrt{n}$ · bootstrap SD of  $\bar{d}_k$
- **Tasks**
  1. RC (White (2000)):  $p = \text{share of replications with } \max_k \sqrt{n}(\bar{d}_k^* - \bar{d}_k) \geq \max_k \sqrt{n} \bar{d}_k$ .
  2. SPA (Hansen (2005)): studentised, recentring only rules with  $t_k \geq -\sqrt{2 \ln \ln n}$ ; StepM (Romano & Wolf (2005)) at 5%: reject every rule above the 95% quantile of the max over the active rules, repeat until none.
  3. Report per benchmark: the largest studentised mean, both  $p$ -values, the number of nulls rejected; plot the bootstrap null of the maximum.
- **Interpretation**
  - ▶ which method handles correlated trials without choosing  $N$ ?
  - ▶ does any rule beat holding Bitcoin?

## B14 [Proposed]: Double Selection: Does the VIX Predict Bitcoin? — Task

- ▣ **Model:** the lecture example “Does the VIX Predict S&P 500 Returns?”
- ▣ **Caveat first:** the plug-in penalty assumes independent scores; with overlapping 5-day outcomes the interval below is an illustrative approximation, not a validated 95% CI
- ▣ **Data:** the modelling frame ( $n = 4,181$ );  $y_t$  = 5-day forward BTC log return in bp;  $d_t$  = log VIX close of the same calendar date, carried forward over weekends and US holidays (the last close known at the BTC close), standardised; controls: the 11 features, their squares and pairwise products ( $p = 77$ ), standardised
- ▣ **Penalty (Belloni, Chernozhukov & Hansen (2014)):**  $\min_{\beta} \frac{1}{n} \sum_t (y_t - \mathbf{x}_t^\top \beta)^2 + \frac{\lambda}{n} \sum_j \hat{\psi}_j |\beta_j|$ ,  $\lambda = 2c\sqrt{n} \Phi^{-1}(1 - \gamma/(2p))$ ,  $c = 1.1$ ,  $\gamma = 0.1/\ln n$ , loadings  $\hat{\psi}_j$  iterated; supplied as `rlasso`
- ▣ **Tasks:** (1) LASSO of  $y_t$  on  $(d_t, \mathbf{x}_t)$ : is the VIX kept? (2)  $\hat{S}_y$  and  $\hat{S}_d$ ; (3) OLS of  $y_t$  on  $d_t$  and  $\mathbf{x}_{\hat{S}_y \cup \hat{S}_d}$  with HAC errors (4 lags):  $\hat{\theta}$ , SE, interval; also OLS with the VIX alone and with all 77 controls
- ▣ **Interpretation:** why does the single LASSO give no valid inference? Read  $\hat{\theta}$  as a predictive coefficient, not a causal fear premium. Notebook: section B14

## C1 [Proposed]: Does a Foundation Model Beat Boosting? — Project Brief

- **Question:** does a time-series foundation model of [Chapter 14](#), used zero-shot, forecast the 5-day direction of Bitcoin better than gradient boosting and a constant?
- **Protocol** (a team project, graded on protocol and evidence, not on the sign of the result)
  1. Name the model and checkpoint, its documented pre-training cut-off, and test only on later dates; otherwise state contamination as a limitation.
  2. Forecast the distribution of the 5-day return directly (not summed daily quantiles);  $P(r_{t,t+5} > 0)$  by linear interpolation of the predictive distribution function at 0, stated tail treatment.
  3. Refit gradient boosting and the constant at every origin on a fixed rolling window (e.g. 3 years); score log losses on identical dates; Giacomini–White test ([Giacomini & White \(2006\)](#)) with HAC errors; a comparison chart.
  4. Optional: a long/cash rule on  $P(\text{up}) > 0.5$  with costs, and its DSR over all configurations and prompts tried.
- **Deliverable:** a Quantlet folder (code, `Metainfo.txt`, one chart) and one slide with the conclusion and its main limitation

## C3 [Proposed]: A No-Arbitrage SDF on Deciles — Project Brief (1/2)

- **Question:** does a linear no-arbitrage stochastic discount factor (SDF) on characteristic-sorted portfolios beat the Fama–French tangency portfolios ([Fama & French \(2015\)](#)) out of sample, as in the lecture case study ([Chen, Pelger & Zhu \(2024\)](#))?
- **Data:** value-weighted decile portfolios of 15 univariate sorts of the [Kenneth French Data Library](#) (size, book-to-market, operating profitability, investment, earnings/price, cash flow/price, dividend yield, momentum, short- and long-term reversal, accruals, market beta, net share issues, variance, residual variance), monthly, minus the one-month bill rate; the FF3 and FF5 factor files
- **Legs:** per characteristic, the mean of deciles 1–5 and of deciles 6–10: 30 factors  $\tilde{F}_{t+1}$ ; months with all 30 legs and FF5 only, no imputation; save the frozen panel with its download date
- **Limitation:** halves of value-weighted deciles replace the rank-weighted legs of 46 characteristics on individual stocks: an extension on portfolio data, not a replication of the stock-level GAN
- **Split:** estimate on 1967–1986, tune on 1987–1991, test on 1992–2016 and on 2017 to the latest month

## C3 [Proposed]: A No-Arbitrage SDF on Deciles — Project Brief (2/2)

▣ **Model:** A5 (Sharpe ratio inference) and B12 (predictive tests) [Solved]

▣ **Stages**

1. Reproduce the FF3 and FF5 benchmarks: tangency weights  $w \propto \hat{\Sigma}^{-1} \hat{\mu}$  on 1967–1986, monthly SR on 1992–2016 with a 95% CI (Lo (2002)); compare with the lecture (FF5 0.22, FF3 0.20).
2. With  $\mu = \text{mean of } \tilde{F}$  and  $Q = \text{mean of } \tilde{F}\tilde{F}^\top$  (a second moment, not a covariance) on 1967–1986:  $\hat{\theta}_{LS} = Q^{-1}\mu$  (solve the system, do not invert),  $\hat{\theta}_{EN} = \arg \min_{\theta} \|\mu - Q\theta\|_2^2 + \lambda_2 \|\theta\|_2^2 + \lambda_1 \|\theta\|_1$  over a reported grid, chosen by SR on 1987–1991.
3. With weights fixed: SR of  $F = \hat{\theta}^\top \tilde{F}$  on both test periods; paired SR differences on aligned months (HAC delta method or block bootstrap); the DSR over the grid.
4. Loadings  $\hat{\beta}_i = \overline{R_i^e F}$  on 1967–1986; project test returns on  $\hat{\beta}$  each month; explained variation  $1 - \overline{\hat{\epsilon}^2} / \overline{(R^e)^2}$  and cross-sectional  $R^2$  per characteristic; a chart of realised vs. implied mean returns.

▣ **Interpretation**

- ▶ why can a linear no-arbitrage SDF beat a forecasting network (Chen, Pelger & Zhu (2024), Table I)?
- ▶ does your post-2016 result survive the configurations tried?

## References (1/5)

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